

Comparative Analysis to Predict Breast Cancer using Machine Learning Algorithms: A Survey

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Abstract—Breast Cancer is the second most dangerous cancer in the world. Most of the women die due to breast cancer not only in India but everywhere in the world. In 2011, USA stated that one in eight women suffered from cancer. Breast cancer develops due to the abnormal cell division in the breast itself which results in the formation of either benign or malignant cancer. So, it is very important to predict breast cancer at an early stage and by providing proper treatment, many lives can be saved. This paper aims to give a comparative study by applying different machine learning algorithms such as Support Vector Machine, K-Nearest Neighbour, Naïve Bayes, Decision Tree, K-means and Artificial Neural Networks on Wisconsin Diagnostic dataset to predict breast cancer at an early stage.

Keywords— Artificial neural network, breast cancer, support vector machine, human disease

I. INTRODUCTION

Cancer which develops in the breast is called Breast Cancer and is the second most dangerous cancer in the world. It is responsible for the death of many women around the globe. More than thirteen thousand Indians die every day due to cancer, according to the National Cancer Registry Programme of the India Council of Medical Research (ICMR). Between 2012 and 2014, the mortality rate due to cancer increased by approximately 6% [1]. Most doctors do the biopsy in order to check whether the patient has cancer or not and whether it is benign or malignant. Cancer that is benign can be said to be “non-cancerous cancer” as it does not spread to other parts of the body whereas a malignant cancer is fatal as it spreads throughout the body and is uncontrollable.

The cure to cancer has not been found yet. The only way to save a person’s life suffering from it is to remove that portion of the body that has been affected. The best way

is to save a person from even developing cancer, and that can be achieved by a timely diagnosis.

Machine Learning is a technique that can learn and retrieve information from data and use that ‘gained’ experience to predict the required outcomes. Machine Learning algorithms have provided great assistance in many fields and early stage cancer prediction.

In computer science, machine learning can be classified in three different ways as supervised learning, unsupervised learning and reinforcement learning. In supervised learning, labelled data is available, and based on that labelled data, the machine predicts the label of the unlabelled input features, whereas, in unsupervised learning, all the features are available without any label or output class. In the case of reinforcement learning, it is the technique of letting the models learn on their own. In this type of learning the machine performs a specific task and either rewards or penalizes itself based on a set of defined rules, and mainly focuses on maximizing the total reward.

There are several machine learning techniques that can help to predict whether the person affected is having a benign or malignant cancer, and this process would be efficient and without any errors. In this paper, authors used a total of six machine learning algorithms to predict the breast cancer.

The further sections are as follows: section 2 describes the existing literature work done by other researchers. Section 3 explains the methodology of the implemented work. Section 4 describes the used dataset with experiments results. Section 5 concludes the work with future scope.

II. RELATED WORK

Cancer is the most dangerous disease in the world. There are different types of cancer such as lung cancer, breast cancer, brain cancer, sarcoma cancer, carcinoma cancer, etc. Out of these many cancers, breast cancer is the second most dangers of cancer in terms of death ratio. If

there is no proper treatment/diagnosis on time, it tends to be quite fatal and lead to death. Many researchers did research to predict breast cancer by applying different machine learning algorithms as Decision Tree (DT) [2], Neural Network (NN) [3], Random Forest (RF) [4], Naïve Bayes (NB) [5], Linear Regression (LR) [6], Support Vector Machine (SVM) [7], and many more on different data set. Table 1 shows the existing literature

work done by researchers. In table 1, column 1 includes the year and author name. Column 2 shows the name of algorithms used by researchers, column 3 includes the accuracy of used algorithms. Column 4 and column 5 shows the advantage and disadvantage respectively of the applied techniques.

TABLE1. COMPARATIVE LITERATURE REVIEW OF EXISTING ALGORITHMS

<i>Year & Authors</i>	<i>Algorithms Used</i>	<i>Accuracy (%)</i>	<i>Advantage</i>	<i>Disadvantage</i>
Adel Aloraini, 2012	Bayesian	97.2,	It works well with data that has highly dependent attributes	It works only with data that has two outcomes to predict
	Network, NN, DT	95.58, 95.49		
Vikas Chaurasia , Saurabh Pal, 2014	Naïve Bayes, RBF Networ k, J48	97.36, 96.77, 93.41	They are highly scalable.	These classifier makes a very strong assumption on the shape of training data distribution
Peter Adebayo Idowu, Jeremiah Ademola Balogun, Kehinde Oladipo Williams and Adeniran Ishola Oluwaranti 2015	Naïve Bayes', J48 Decision Trees	82.6, 94.2	Easy to handle noisy data.	Large number of training dataset required for proper training.
Thomas Noel, Hiba Asri, Hajar Mousannif, Hassan Al Moatassime, 2016	C4.5, SVM, NB, K-NN	95.13, 97.13, 95.99, 95.27	Risk of overfitting is less	Best parameters are needed for correct classification
Yixuan Li, Zixuan Chen, 2018	DT, SVM, RF, LR, NN	96.1, 95.1, 96.1, 93.7, 95.6	It runs efficiently on large databases	They are much harder to construct and are time taking
Priyanka Gupta, Prof. Shalini L, 2018	CART , RF, K-NN, Boosted Trees	92.35, 96.47, 97, 96.47	It is very robust and can be simply implemented on classification datasets	It does not learn anything from the training set and just uses it to predict the class of the test points
Jabeen Sultana, Abdul Khader Jilani, 2018	LR, MLP, RF, DT	97.18, 95.25, 95.25, 93.14	If all the useful independent variables have been identified and used, then it is a good classifier	Doesn't perform well with a large dataset
Kriti Jain1, Megha Saxena, Shweta Sharma. 2018	SVM	97.13	Risk of overfitting is less	Best parameters are needed for correct classification
Puneet Yadav, Rajat Varshney, Vishan Kumar Gupta, 2018	Decision Tree, SVM	90.29, 94.5 - 97	DT require little effort for their preparation	They are unstable and complex to understand when there are many outcomes

III. METHODOLOGY

A. Data Collection

To predict breast cancer, authors used Wisconsin Diagnostic dataset [8] collected from the UCI Machine Learning Repository. There were 699 instances with total of eleven features. Out of eleven features, ten features as input features and remaining one feature is treated as output feature. The whole dataset is divided into training and testing instances in the ratio of 80:20. It means out of 699 instances, 560 instances were used as training dataset and the remaining 140 instances were used as a testing dataset.

B. Data Pre-processing

Wisconsin Diagnostic dataset for breast cancer prediction has some missing values. To handle these missing values data pre-processing was also used on the mentioned dataset. The attribute like Bare Nuclei column has missing features in the form '?' string which need to be inputted. 16 such instances of missing values were found in this feature. These missing values were replaced by the average/mean values of the features. On the other hand, the attributes like sample code number have no relevance in predicting breast cancer so such types of attributes have been dropped from the dataset.

C. Algorithm Used

Support Vector Machine: Support vector machine (SVM) is a quite simple classification algorithm. This classifier is named so because it takes the help of vectors in the feature space to classify the class of a new vector [9,10]. The Maximum Margin Hyper-plane (MMH) decides whether the new vector belongs to class one or class two. If the data point lies beyond the negative hyper-plane or to the left of MMH then it belongs to the class one, else it belongs to the class two, where class one and two are two different classes in a given situation. SVMs can also be used if there are more than two classes.

K-Nearest Neighbour: K-Nearest Neighbour (KNN) is said to be the simplest and the most straightforward classification algorithm. Like most machine learning algorithms, K-NN does not learn anything from the provided dataset and its attributes, but simply use the points from the training data and finds the K number of nearest neighbours to that data point using Euclidean Distance [11] and classify it to the class which has the first K neighbours closest to it.

Naïve Bayes: Naïve Bayes (NB) theorem is a machine learning algorithm that works on the probability concepts as mentioned in equation 1.

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)} \quad (1) \quad (2)$$

Where P(A) is the Prior Probability, P(B) is the Marginal Likelihood, P(B|A) is the Likelihood and P(A|B) is the Posterior Probability. The NB algorithm follows the above equation for the determination of the class of a data point. The posterior probability is calculated based on the position of the vector in the feature space and then the data point is assigned to the class with greater posterior probability [12].

Decision Tree: Decision Tree (DT) is a powerful machine learning algorithm used for both classification as well as regression [13]. DT can be a tree type of structure where each internal node is a test condition for the vector to move further and the terminal nodes represent the class or the prediction value to be predicted. DT is good for the classification of a few class labels but do not produce proper results if there are many classes and less training observations. And moreover, DTs can be expensive to train computationally.

K-Means: K-Means algorithm is generally used for clustering, but it can also be used for classification, as making a cluster of similar data points is equivalent to classifying similar data points to their respective classes. The very first step in the working of this algorithm is to determine the number of clusters, and this determined by a method called The Elbow Method [14]. The number of clusters is represented by 'K'. Next, it works in a unique way by randomly initialising 'K' centroids in the feature space and then use Euclidian distance to find the nearest data points to the respective centroids. Then the centroids

are moved a bit randomly and this process repeats until there are no more reassignments of the data points to any centroid. Finally, the clusters are determined, and the process ends. The only drawback of K-Means is the Random Initialisation Trap, that leads to the formation of wrong clusters, overcome it by selecting the 'K-Means++' as the random initializer.

Artificial Neural Network: Artificial Neural Network (ANN) is one the most advanced and powerful machine learning algorithm that can be used for many purposes such as classification, regression, voice recognition, targeted marketing, etc. A single unit of an ANN is called a perceptron, it was first invented in 1957 by Frank Rosenblat [15], as he wanted to invent something that could learn and adjust itself to meet the required needs.

The way in which ANNs work is a bit complex but understandable, firstly the input nodes are taken and the number of input nodes is equal to the number of independent variables, then these inputs are weighted through synapses and passed on to the next neuron where a Activation Function is applied and this process repeats till the weights have reached the Output Layer. The output is taken as $\hat{y}$ and the actual value is taken as y . Now the Cost Function (C) is calculated according to the equation 2.

$$C = \frac{1}{2}(\hat{y} - y)$$

The aim of the ANN here is to minimize the cost function and this is achieved by the process of Back Propagation. If the value of cost function is high above a certain level, then the information is fed back into the Neural Network and the weights are modified and this process is repeated till the C function is minimized. This flow of the ANN is repeated for every observation in the dataset and finally the ANN gets ready for real world classification.

Figure 1 shows the used architecture of ANN. Authors uses ten neurons as input layer and two hidden layer each of six neurons. At the end of mentioned architecture, only single neuron used because to predict patient suffer from breast cancer or not.

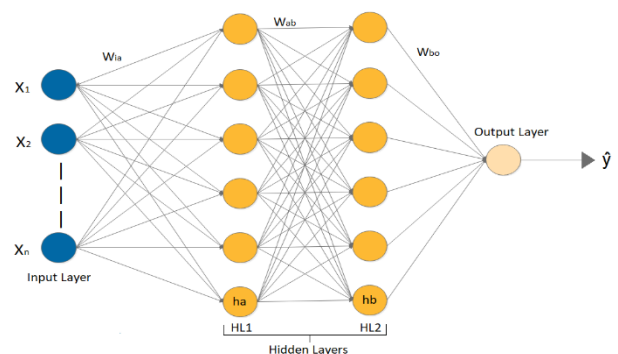


Figure 1. Architecture of used Artificial Neural Network

IV. EXPERIMENT RESULTS

A. System Specification

The model is trained on the Google Colab, which is a free online training platform. Google Colab uses Tesla K80 GPU and provides a RAM of 12 GB and 12 hours of continuous use. Due to RAM limitations the batch size is kept small.

B. Results and Discussion

To predict the breast cancer, authors apply different machine learning algorithms as SVM, KNN, NB, DT, ANN etc. The applied models were compared based on the Precision (P), Recall (R) and accuracy which can be calculated using equation 3, 4 and 5 respectively.

$$P = TP / (TP + FP) \quad (3)$$

$$R = TP / (TP + FN) \quad (4)$$

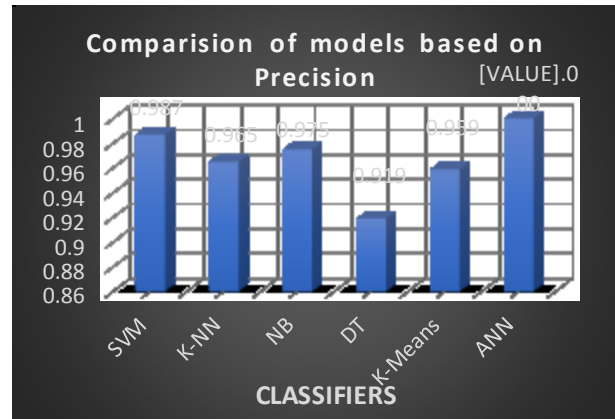
$$\text{Accuracy} = (TP + TN) / \text{Total No of Instances} \quad (5)$$

Where, TP and TN stand for True Positive and True Negative respectively, and FP and FN stand for False Positive and False Negative respectively.

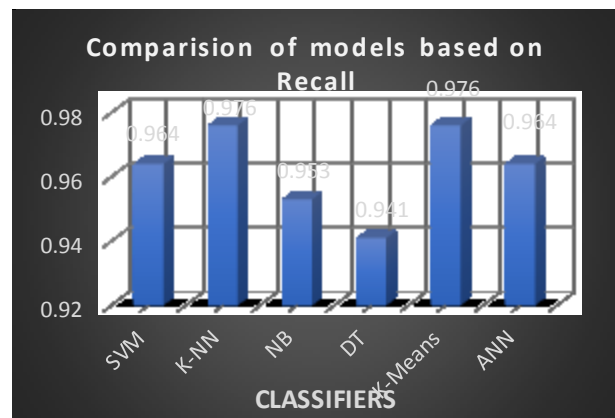
Table 2 shows the result analysis of the used techniques. In table 2, column 1 includes the name of techniques used to predict breast cancer. Based on the Confusion matrix parameters such as TP, TN, FP, FN, Column 6 and 7 shows the precision and recall value of the applied techniques, Column 8 includes the accuracy of used algorithms. The comparison of applied techniques in terms of Precision, Recall and Accuracy are shown in Figure 2.

TABLE2. RESULT ANALYSIS OF APPLIED TECHNIQUES

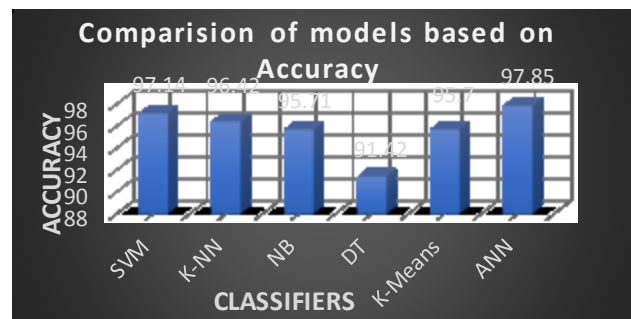
Techniques	TP	FP	TN	FN	P	R	Accuracy (%)
SVM	82	1	54	3	0.987	0.964	97.14
K-NN	83	3	52	2	0.965	0.976	96.42
NB	81	2	53	4	0.975	0.953	95.71
DT	80	7	48	5	0.919	0.941	91.42
K-Means	446	19	22	11	0.959	0.976	95.70
ANN	82	0	55	3	1.000	0.964	97.85



(a)



(b)



(c)

Figure 2. Comparison of models based on a) Precision b) Recall c) Accuracy

V. CONCLUSION AND FUTURE WORK

Breast Cancer is fatal and needs early detection in order to be cured. This year, an estimated 268,600 women in the United States will be diagnosed with invasive breast cancer, and 62,930 women will be diagnosed with in situ breast cancer. An estimated 2,670 men in the United States will be diagnosed with breast cancer. So, prediction techniques are necessary to achieve the detection of cancer. To predict the breast cancer in an advanced stage, authors applied six different machine learning algorithms as DT, NB, LR, RF, SVM, and NN on Wisconsin Breast Cancer dataset which is publicly available on the internet. After comparing all the applied algorithm, the authors conclude

that artificial neural network gives better prediction as 97.85% compared to all other algorithms. The best accuracy is given by ANN and hence can be used to predict cancer and save lives at present as well as in future. This accuracy can be improved by increasing data size in future.

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